

Presentation Script

Vocabulary Learning Support System Using LLMs and Video Generation AI

Hello, everyone.

Today, I will talk about my research. The title is "Vocabulary Learning Support System Using LLMs and Video Generation AI."

First, I will explain the background. Engineering students need English vocabulary for papers, technical documents, and tests. However, learning with a word book is often difficult. One reason is that words are separated from real usage scenes. If we can connect words with scenes and experiences, students may remember them more easily.

The purpose of this research is to support English vocabulary learning. In this study, I try to connect English words with concrete usage scenes. I use large language models and video generation AI to create learning episodes automatically.

Next, I will explain the system flow. First, a learner enters a target English word. Then, an LLM generates a natural example sentence based on the meaning of the word. Next, another LLM step changes the sentence into a video prompt. The prompt describes the scene for video generation. After that, video generation AI creates a short video. The learner studies the word, the example sentence, and the video together.

The system shows three things: the target English word, an example sentence, and a short video of the usage scene. By using these three items, the learner can connect the word with a memorable episode.

At this stage, I am organizing the basic system design. I have also tested video generation with some English words. Concrete action words are relatively easy to visualize. For example, words about actions or situations can be shown in a video. On the other hand, abstract words are still difficult to express. This is one of the main challenges of my research.

The final goal is to make vocabulary learning more memorable than word-book learning.

Thank you for listening.